Project Defense Study Guide

Order Management System with Sales Tracking and Report Generation for Aling Chona Cakes and Cupcakes

This guide contains nine ERD questions, 15 core panel questions, and 12 tougher defense questions with suggested answers. Practice answering each question in your own words before reviewing its answer.

Audio companion: Microsoft Zira. Duration: 19 minutes 45 seconds. The 27 core and tougher questions follow the recording, with their start times shown beside each question. The ERD questions are additional reading material.

ERD questions

ERD question 1

Why did you change the ERD format instead of following the sample exactly?

Suggested answer. We adapted the layout because our system has 23 business tables and 40 foreign-key relationships. Putting all attributes and connections into one diagram would make it difficult to read. We retained the table style, keys, and cardinalities while presenting the relationships more clearly.

ERD question 2

Why did you divide the ERD into seven detailed views?

Suggested answer. We grouped related tables by function, such as orders, payments, and inventory. This allows readers to examine each part of the system clearly. The overview shows how the main entities connect across these functions.

ERD question 3

Why do some tables appear in several diagrams? Does that mean your database contains duplicate tables?

Suggested answer. No. Repeated tables represent the same database entities. For example, the users table appears in several views because staff members are associated with different transactions. We repeated it so readers can understand each view without switching pages.

ERD question 4

How can you prove that splitting the diagram did not leave out relationships?

Suggested answer. We checked the detailed views against the schema and data dictionary. Together, they cover all 23 documented business tables and all 40 declared foreign-key relationships. Each repeated table also uses the same attributes and key labels.

ERD question 5

Give us an example of a one-to-many relationship in your system.

Suggested answer. Orders and order details have a one-to-many relationship. Multiple order-detail records can reference the same order through order_id, while each order-detail record references one order. This allows an order to contain multiple items.

ERD question 6

Why are some relationships optional?

Suggested answer. Some related records are only assigned when a particular action occurs. For example, an order's reviewed_by field can be empty before a staff member reviews it. Its relationship to users is therefore optional, based on the foreign key's nullable constraint.

ERD question 7

Why do you have junction tables such as add_on_product?

Suggested answer. A product can have multiple available add-ons, and an add-on can be associated with multiple products. The junction table records these associations using their keys, allowing us to represent the many-to-many relationship without adding repeated columns.

ERD question 8

Why did you omit fields such as created_at and version?

Suggested answer. We focused the paper's ERD and data dictionary on the documented business attributes and relationship keys. Routine tracking and implementation fields were omitted for readability. This makes the diagram a focused representation of the business database rather than a complete listing of every implementation column.

ERD question 9

How do you ensure the ERD and data dictionary are consistent?

Suggested answer. We matched every detailed table's attribute names, order, and key labels to the data dictionary. We also checked the relationships and cardinalities against the schema constraints. This ensures that the two sections describe the same documented database design.

Core panel questions

Question 1 **Audio 00:34**

What specific problems does your project address?

Suggested answer. We identified three main problems: scattered order information across Messenger conversations and handwritten records, insufficient sales and expense reporting, and improper inventory tracking. These make it difficult to monitor orders, understand financial activity, and anticipate supply shortages.

Question 2 **Audio 01:09**

How did you determine that these were the business's actual problems?

Suggested answer. We gathered information through an initial interview with the business owner and examined the existing workflow. The owner described difficulties with tracking orders, understanding sales and expenses, and monitoring supplies. These findings became the basis of our requirements.

Question 3 **Audio 01:41**

How does your proposed system address those problems?

Suggested answer. It centralizes customer orders, customization details, payments, and pickup schedules. It also organizes sales and expense information, generates reports, and records inventory movements. These functions reduce the need to search through separate conversations and handwritten records.

Question 4 **Audio 02:14**

Why did you choose the Waterfall methodology?

Suggested answer. We chose Waterfall because its defined phases helped us organize requirements, design, implementation, and testing within the academic timeframe. Gathering the business requirements early gave us a clear foundation for development. Deployment and maintenance remain subsequent activities.

Question 5 **Audio 02:46**

Is this system practical for a business with only two staff members?

Suggested answer. Yes, it is designed around the Owner and Assistant's existing responsibilities. Browser access allows them to use familiar devices, and open-source tools reduce software licensing costs. However, hosting arrangements, training, and ongoing maintenance still need to be addressed.

Question 6 **Audio 03:20**

Why must staff confirm an order before the customer pays?

Suggested answer. Customized cakes require staff to check whether the requested design and pickup schedule are feasible. After confirmation, the customer can proceed with the required fifty percent down payment. Production requires a verified deposit, which helps prevent work from beginning on an unconfirmed request.

Question 7 **Audio 03:54**

Does your system automatically verify G Cash payments?

Suggested answer. No. The customer submits payment proof, but the Owner must check the actual transaction outside the application before recording it as verified. Uploading a receipt alone does not establish that the business received the payment.

Question 8 **Audio 04:23**

Why do your reports distinguish sales from payment collections?

Suggested answer. They represent different events. A deposit can be collected before an order is completed, while completed-order sales represent fulfilled orders. Separating them prevents counting the same transaction twice. Our operational result is completed sales plus retained cancellation deposits minus valid expenses. It is not full accounting profit.

Question 9 **Audio 05:02**

How does inventory management solve the problem if staff still enter transactions manually?

Suggested answer. It provides organized records of received supplies, usage, waste, adjustments, and current balances, together with low-stock warnings. This improves visibility compared with relying on observation alone. However, accuracy still depends on timely entries and physical stock checks.

Question 10 **Audio 05:38**

Why did you not implement automatic ingredient deductions?

Suggested answer. Reliable deductions require accurate ingredient quantities and standardized recipes for each product or variation. Our current system records actual stock movements entered by staff. Recipe-based deductions are identified as a possible future improvement once the necessary measurements are available.

Question 11 **Audio 06:11**

Why do the Owner and Assistant have different permissions?

Suggested answer. Their access follows their responsibilities. The Owner controls payments, cancellations, accounts, catalog settings, and financial reports. The Assistant handles authorized operational tasks, including feasibility confirmation, production-status updates, inventory, and expenses. These restrictions are enforced on the server.

Question 12 **Audio 06:48**

Your implementation plan mentions deployment. Was the system actually deployed?

Suggested answer. Public deployment was planned but was not completed during the project period because hosting costs and arrangements had not been finalized with the Owner. The completed features were tested locally and demonstrated through Google Meet on October ninth, twenty twenty-six.

Question 13 **Audio 07:21**

Why did only the Owner and Assistant evaluate the system? Is that sufficient?

Suggested answer. They are the intended staff users, so their feedback is directly relevant to the business workflow. However, two evaluators provide an initial assessment. Their responses cannot establish customer usability, long-term reliability, or performance during normal business operations.

Question 14 **Audio 07:56**

What do your high evaluation ratings actually prove?

Suggested answer. The paper reports overall means of four point nine zero for usability, four point nine eight for functional requirements, and four point eight nine for non-functional requirements. These show positive feedback from the two staff evaluators. They do not independently prove technical security, compatibility across all devices, or success in live operation.

Question 15 **Audio 08:35**

What important improvements emerged from the client's feedback, and what was your contribution?

Suggested answer. The feedback highlighted the need for customer order-modification requests and less manual inventory work. Future work also includes public deployment and customer testing. My contribution covered overall system development, the database and modules, and documentation, including checking their consistency with the business requirements.

Tougher defense questions

Question 16 **Audio 09:19**

You chose Waterfall, but your project changed during development. Does that contradict your methodology?

Suggested answer. Waterfall gave us a structured sequence for gathering requirements, designing the system, implementing features, and testing them. Revisions still required us to revisit the affected requirements, design, and checks. We should describe those revisions honestly instead of claiming that the first specification never changed. We also distinguish the phases completed during the project from public deployment and continuing maintenance, which remain subsequent work.

Question 17 **Audio 10:08**

If the system was not publicly deployed, how can you defend calling the project completed?

Suggested answer. We can defend completion of the documented development deliverable: the implemented features, local testing, demonstration, and paper. We cannot claim that public rollout or validation in daily business operations has been completed. Hosting arrangements, staff preparation, and live evaluation remain necessary. Our conclusion should stay within the evidence from the local implementation and the client demonstration.

Question 18 **Audio 10:52**

With only two evaluators and very high scores, how do you address possible bias in your evaluation?

Suggested answer. The Owner and Assistant are the actual intended staff users, so their feedback is useful for checking whether the workflow fits the business. However, the sample is very small, and a guided demonstration may influence responses. We treat the scores as initial user feedback rather than a representative statistical result. A stronger evaluation would include actual customers, independent reviewers, and observation during normal business use.

Question 19 **Audio 11:38**

Your security rating is five. Does that prove that the system is secure?

Suggested answer. No. That rating reflects the evaluators' perception during the demonstration. It does not replace a technical security assessment. The paper documents safeguards such as password hashing, server-side permissions, input validation, and private storage for payment receipts. We can explain those controls, but an independent security review and focused testing would provide stronger evidence. We should not describe the questionnaire score as proof that vulnerabilities do not exist.

Question 20 **Audio 12:28**

What evidence proves that your system improved efficiency if you did not measure the previous process?

Suggested answer. The demonstration showed that related order information and reports can be accessed through one system. That supports the practical value of centralizing records. However, our paper does not provide a before-and-after measurement of processing time, errors, or profit. We should avoid claiming a specific percentage improvement. Future evaluation can compare the time and accuracy of the same tasks using the previous workflow and the deployed system.

Question 21 **Audio 13:14**

Your paper discusses updating inventory balances, but you also say inventory updates are not automatic. Which statement is correct?

Suggested answer. The balance changes when staff record a stock movement, such as receiving ingredients, recording usage, or making an adjustment. The system does not calculate ingredient consumption from an order and deduct it automatically when that order is completed. These are different processes. Our documentation should clearly connect inventory balance updates to the transactions entered by staff so that readers do not assume recipe-based automation is already implemented.

Question 22 **Audio 14:05**

What happens if staff forget to record ingredient usage? Can you still trust the low-stock warning?

Suggested answer. If a movement is missing, the recorded balance can be higher than the actual stock, and a warning may come too late. The system organizes information, but it cannot observe physical consumption on its own. Timely transaction entry, clear staff responsibility, and regular physical counts are therefore necessary. When a difference is found, staff should investigate it and record an appropriate adjustment. This limitation remains until reliable recipe data or another consumption-recording method is available.

Question 23 **Audio 14:57**

Customers access their orders through private links. What if a link is shared or leaked?

Suggested answer. A private link must be treated as confidential because someone who obtains a valid link may be able to access the associated customer order page. Random links, exclusion from indexing and caching, and private receipt storage are documented safeguards, but they do not remove the risk of disclosure by the customer. We should explain this limitation honestly. Stronger access controls, such as expiry or an additional verification step, could be assessed as future improvements rather than presented as existing features.

Question 24 **Audio 15:48**

If the Assistant opens a restricted address directly, can hidden buttons alone prevent unauthorized actions?

Suggested answer. Hidden buttons alone are insufficient. Authorization must also be checked on the server whenever a protected page or action is requested. Our system uses role restrictions so that the Assistant cannot gain the Owner's payment-verification or financial-report access simply by entering an address. A useful defense demonstration would show an allowed operational action and a denied restricted request. The explanation should focus on server enforcement, not only on what the interface displays.

Question 25 **Audio 16:38**

Why build a custom system instead of continuing with Messenger, spreadsheets, or a generic point-of-sale application?

Suggested answer. Our requirements include customized cakes, product layers and inclusions, design notes, reference images, feasibility confirmation, deposits, and pickup schedules. A custom system allows these details to follow one consistent workflow with separate Owner and Assistant permissions. This explains the design decision, but we did not perform a comprehensive comparison of every commercial alternative. We should defend the fit to the documented requirements without claiming that other tools could never meet them.

Question 26 **Audio 17:31**

Why did you present the entity relationship diagram as an overview and several detailed views instead of one large diagram in the given format?

Suggested answer. The database has twenty-three business tables and forty declared foreign-key relationships. Putting every table, attribute, and connector into one figure would make the printed diagram difficult to read. We kept an overview for the overall structure and provided seven focused detailed views in the appendix. The detailed views use the same field names and key labels as the data dictionary, and together they cover the declared relationships. A table repeated in another view represents the same database table, not a duplicate table. The purpose of this presentation is to make the design readable and traceable. We should still acknowledge the format adjustment and explain it to the panel.

Question 27 **Audio 18:35**

Your data dictionary and diagram omit fields such as creation timestamps and version values. Does that make the documentation incomplete?

Suggested answer. They are presented as a business-focused description of the database rather than an exhaustive physical schema listing. We retained the documented business attributes and the keys needed to explain the relationships, and aligned the diagram with the data dictionary. Routine implementation fields were omitted to keep the paper readable. We should state that scope clearly instead of claiming that every physical column appears in the paper. The current database schema remains the source for technical details that were intentionally excluded.

For your defense, connect each answer to the business problem, the implemented feature, and the evidence in your paper. Be clear about what was completed, what was evaluated, and what remains future work.